

Lecture 4.2: Reporting the Conceptual Framework

AREC 570 / ECON 530 · Unit 4

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Readings and sequence

1. Chouinard, “Theory Notes”

- What belongs in the theory section of an empirical paper

2. Cheng, “How to Write Proofs: A Quick Guide”

- How to make each step in an argument follow from what precedes it

3. Application

- How to report the SNAP retailer framework developed in Lecture 4.1

Start with the claim

Because stricter stocking standards require healthier foods, they improve food access and diet quality.

Before accepting the claim, ask

- Which conclusion follows from the model
- Which outcome is observed in the data
- Which links require additional evidence

The framework may explain changes in retailer authorization. The available data do not establish changes in inventory, purchases, diet, or health.

From constructing the framework to reporting it

Lecture 4.1

Define the research problem

Identify agents choices and constraints

Derive conditional predictions

State what the model and data do not establish

Lecture 4.2

Explain why the problem requires theory

Introduce each element in a readable order

Show how each conclusion follows

Keep the written claim within those boundaries

Lecture 4.1 asked what the framework can support. Lecture 4.2 asks how a reader can see why.

Chouinard

Theory has a specific job

Chouinard argues that every paper has a theoretical foundation whether or not the author makes it explicit ([Chouinard 2015](#)).

For an empirical paper the theory section should

- Explain the mechanism behind the research question
- State the assumptions needed for that mechanism
- Derive the hypotheses evaluated with data
- Connect theoretical concepts to empirical measures
- Bound the conclusions the evidence can support

The role of theory depends on the contribution

Theoretical paper

Develop or modify a model

Establish analytical results

Show what changes relative to prior theory

Treat the model as a central contribution

Empirical paper

Apply theory to explain the empirical question

Derive hypotheses and comparative predictions

Connect concepts to observed measures

Use the model to discipline the empirical argument

Both require explicit assumptions and a coherent sequence of reasoning.

Source: Chouinard (2015)

Match the detail to the theoretical contribution

Apply established theory

Summarize the relevant result and explain how it applies to the research problem

Modify established theory

State the baseline model identify the change and show how the result changes

Develop new theory

Define the setting assumptions relationships and results in sufficient detail to establish the contribution

The amount of theory should reflect its role in the paper.

Build from infrastructure to superstructure

Introduce each component before asking the reader to interpret the completed model.

1. Define the setting and relevant agents
2. Identify choices and constraints
3. State the maintained assumptions
4. Introduce the relationships among model objects
5. Derive and interpret the result
6. Connect the result to the empirical question

Source: Chouinard (2015)

Apply that sequence to the SNAP example

Setting and agents SNAP-authorized retailers respond to a revised stocking standard

Choice Each retailer decides whether authorization remains worthwhile

Constraint Continued authorization may require costly changes in qualifying inventory

Result Some retailers may lose authorization when compliance cost exceeds the value of participation

Empirical question How did the number type and location of authorized retailers change

Keep definitions assumptions and results distinct

Element	SNAP example
Definition	$a_i(r) = 1$ means retailer i is authorized under standard r
Maintained assumption	The stronger standard raises compliance cost for at least some retailers
Model result	A retailer may lose authorization when compliance cost exceeds its value

Definitions assign meaning. Assumptions establish the conditions for the reasoning. Results must follow from both.

Use notation only when it reduces the reader's burden

For retailer i

- r is the strictness of the stocking standard
- B_i is the value of SNAP authorization
- $K_i(r)$ is the cost of meeting standard r
- $a_i(r)$ is authorization status

The index i matters because retailer values and costs differ. Additional notation should be introduced only when it changes or clarifies the argument.

Make the equation readable

The retailer remains authorized when the value of authorization covers the compliance cost

$$a_i(r) = \begin{cases} 1 & \text{if } B_i \geq K_i(r) \\ 0 & \text{if } B_i < K_i(r) \end{cases}$$

In words a higher-cost standard can change authorization only for retailers near enough to the participation threshold.

A model result is not yet an empirical finding

Model result

Holding B_i fixed a higher $K_i(r)$ can cause a retailer to lose authorization

Empirical hypothesis

Retailers expected to face higher compliance costs are more likely to lose authorization after implementation

Evidence

Store-year authorization data can show whether losses differ by observable retailer characteristics

The evidence may support or challenge the hypothesis without revealing the unobserved compliance cost itself.

Discussion: apply Chouinard to your research

1. What role does theory serve in your paper
2. Are you applying modifying or developing a model
3. Which concept must be defined before the reader sees an equation
4. Which assumption carries your principal result
5. What empirical hypothesis follows from that result

Cheng

A defensible argument has a beginning middle and end

Cheng describes a proof as a sequence of statements in which each statement follows from what came before ([Cheng 2004](#)).

Part	Function in an economic argument
Beginning	Definitions assumptions and accepted results
Middle	The steps connecting the premises to the result
End	The claim the argument was constructed to establish

The reader should not have to supply a missing step.

Plan backward and write forward

When developing an argument it is useful to ask

| What would have to be true for this conclusion to follow?

That question works backward from the intended result and helps identify the required assumptions and intermediate steps.

The written argument should then proceed forward from the stated premises to the result.

Apply the beginning middle and end

Beginning

The revised standard raises $K_i(r)$ for some retailers while B_i is held fixed

Middle

A retailer remains authorized only when $B_i \geq K_i(r)$

End

A retailer that crosses the participation threshold loses authorization

Each part is necessary but each part has a different role.

Show the threshold crossing

Suppose the standard changes from r_0 to r_1 and

$$K_i(r_0) \leq B_i < K_i(r_1)$$

Then

$$a_i(r_0) = 1 \quad \text{and} \quad a_i(r_1) = 0$$

The retailer was authorized under the earlier standard and is not authorized under the stronger standard.

Do not take a flying leap

The authorization model can support

higher compliance cost→possible authorization loss

It does not by itself support

authorization loss→diet or health change

The second claim requires additional relationships and evidence about inventory availability purchases and health.

Two other errors to avoid

Incorrect converse

A retailer remained authorized therefore the revised requirement was slack

The retailer may instead have changed its inventory to comply.

Unstated causal assumption

A retailer lost authorization after implementation therefore the revised rule caused the loss

The analysis must distinguish policy exposure from closure reclassification and other changes over time.

Audit these claims

A. A higher compliance cost can reduce authorization when the value of authorization is held fixed

B. A retailer that remains authorized must have faced a slack requirement

C. Authorization losses concentrated in high-SNAP communities show that the rule reduced diet quality

For each claim identify

- What follows from the model
- Which assumption is required
- What additional evidence would be needed

What follows from the argument?

Claim A follows conditionally from the participation model

Claim B does not follow because continued authorization is consistent with either prior compliance or an inventory adjustment

Claim C may identify a targeting concern for retailer access but it does not establish a diet-quality effect

Discussion: apply Cheng to your research

1. What is the final claim your framework is intended to establish
2. Which premises must be stated before that claim
3. Which intermediate step is most likely to be omitted
4. Have you treated a sufficient condition as if it were necessary
5. Where might your prose claim more than the model establishes

Organize the written conceptual framework

1. State the research problem that requires explanation
2. Identify the relevant economic theory
3. Define agents choices constraints and key terms
4. State the maintained assumptions
5. Derive and interpret the model result
6. Translate the result into an empirical hypothesis
7. State what the model and available evidence do not establish

The order should help the reader reconstruct the argument without reverse engineering it.

Connect model objects to available measures

Model object	Empirical counterpart
r	Implementation of the revised stocking standard
a_{it}	Store i authorization status in year t
A_{mt}	Number and composition of authorized stores in market m and year t
S_{mt}	Share of households participating in SNAP

The data do not directly measure B_i $K_i(r)$ or store inventory.

Framework measurement and identification answer different questions

Conceptual framework

Why might the stocking standard change retailer authorization

Measurement

Which observed variables represent authorization retailer type location and program reliance

Identification

What comparison isolates the response to the policy from other changes over time

The conceptual framework informs the design but does not substitute for it.

First draft: where does the argument overreach?

The revised rule increased required inventory and therefore improved food access. Stores unable to meet the rule lost authorization, so we expect fewer authorized retailers and healthier inventories after 2018.

Ask whether the paragraph

- States the maintained assumptions
- Separates model results from empirical hypotheses
- Distinguishes observed from unobserved outcomes
- Limits the conclusion to what the evidence can support

Revised paragraph

The revised standard may raise the cost of continued authorization for retailers that did not already satisfy it. Holding the value of authorization fixed, a retailer loses authorization when the compliance cost exceeds that value. We therefore test whether authorization losses increased after implementation and whether losses varied by store type and local SNAP participation. Because the data do not measure inventory, purchases, or health, these outcomes remain outside the empirical claim.

The revision states the mechanism prediction empirical application and boundary in sequence.

Boundaries strengthen the argument

The framework supports a conditional prediction about retailer authorization.

The data can describe changes in the number type and location of authorized retailers.

The study cannot directly establish

- Which retailers faced a binding inventory constraint
- How store inventory changed
- How household purchases changed
- Whether diet or health improved

Apply the reporting sequence to your research

Draft four connected statements

1. **Mechanism** Theory implies that ____ affects ____ because ____
2. **Condition** This prediction requires ____
3. **Evidence** I observe the relevant concepts using ____
4. **Boundary** These data do not establish ____

Then check whether each statement follows from the preceding one.

A reader should be able to answer six questions

1. What problem does the framework explain
2. Which theory organizes the explanation
3. What do the central terms and symbols mean
4. Which assumptions produce the result
5. What empirical hypothesis follows
6. What remains outside the model or evidence

If one answer is missing the written framework is incomplete.

What the readings contribute

Chouinard identifies the components and order of a theory section

Cheng requires each conclusion to follow from stated premises

Together they turn a model into a written argument whose scope a reader can evaluate.

References

Cheng, Eugenia. 2004. *How to Write Proofs: A Quick Guide*. Department of Mathematics, University of Chicago.
Chouinard, Hayley H. 2015. *Theory Notes*. EconS 529 course notes.

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